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Studierichting: Informatica
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$$z^4 = 8(-1 + i\sqrt{3}) = -8 + 8\sqrt{3}i$$

$$r = \sqrt{(-8)^2 + (8\sqrt{3})^2} = 16$$

$$\theta = \frac{2\pi}{3}$$

$$z = 2 \left(\cos \left(\frac{\frac{2\pi}{3} + 2k\pi}{4} \right) + i \sin \left(\frac{\frac{2\pi}{3} + 2k\pi}{4} \right) \right) = 2 \left(\cos \left(\frac{\pi}{6} + \frac{k\pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{k\pi}{2} \right) \right)$$

$$z_0 = 2 \left(\cos \left(\frac{\pi}{6} + \frac{0 \cdot \pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{0 \cdot \pi}{2} \right) \right) = 2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = \sqrt{3} + i$$

$$z_1 = 2 \left(\cos \left(\frac{\pi}{6} + \frac{1 \cdot \pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{1 \cdot \pi}{2} \right) \right) = -1 + i\sqrt{3}$$

$$z_2 = 2 \left(\cos \left(\frac{\pi}{6} + \frac{2 \cdot \pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{2 \cdot \pi}{2} \right) \right) = -\sqrt{3} - i$$

$$z_3 = 2 \left(\cos \left(\frac{\pi}{6} + \frac{3 \cdot \pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{3 \cdot \pi}{2} \right) \right) = 1 - i\sqrt{3}$$

References